MA1513 Weekly Self Practice Set 5 Answer

1. Let $y_1(t), y_2(t)$ and $y_3(t)$ be three functions in t such that

$$\begin{array}{rcrcrcr} y_1' & = & 5y_1 & - & y_2 & - & y_3 \\ y_2' & = & -y_1 & + & 5y_2 & - & y_3 \\ y_3' & = & -y_1 & - & y_2 & + & 5y_3 \end{array}.$$

(a) Write down the matrix form of the above system of differential equations.

Answer

$$\begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \begin{pmatrix} 5 & -1 & -1 \\ -1 & 5 & -1 \\ -1 & -1 & 5 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.$$

(b) Use MATLAB command to find the eigenvalues and eigenvectors of the underlying matrix in (a), and use them to write down a fundamental set of solutions for the above SDE.

(To get nice values for eigenvectors, use the MATLAB symbolic command: $\gg$ `[P D]=eig(sym(A))`.)

Answer

MATLAB will return the matrices $P = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ and $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix}$.

So the eigenvalues of $\boldsymbol{A}$ are $3, 6$ (repeated) with corresponding eigenvectors $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

So a fundamental set of solutions is given by

$\boldsymbol{y}(t) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t}$, $\boldsymbol{y}(t) = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{6t}$, and $\boldsymbol{y}(t) = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{6t}$.

(c) Write down the general solution of the SDE.

Answer

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{6t} + c_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{6t}.$$

(d) Given the initial conditions $y_1(0) = 3, y_2(0) = 3, y_3(0) = 0$, find the particular solution of the initial value problem for the above SDE. Show your working.

Answer

$y_1(0) = c_1 - c_2 - c_3 = 3$, $y_2(0) = c_1 + c_2 = 3$ and $y_3(0) = c_1 + c_3 = 0$.

We solve the equations to get $c_1 = 2, c_2 = 1$ and $c_3 = -2$.

So the particular solution of this initial value problem is

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t} + \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} e^{6t} - 2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} e^{6t}.$$